

FlowPMO: Operationalizing Multidimensional Developer Productivity with Absolute Benchmark Normalization

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Abstract—Measuring developer productivity remains one of the most contested challenges in software engineering management. Widely adopted proxies — lines of code, commit frequency, and velocity points — are uni-dimensional and susceptible to gaming. This paper presents FlowPMO, an open-source Plotly/Dash dashboard that operationalizes developer productivity across five SPACE-inspired dimensions: Delivery Output, Flow Efficiency, Review Quality, Process Conformance, and Anti-Rework, each normalized against absolute, literature-grounded benchmarks. Beyond the five radar dimensions, FlowPMO introduces three composite indices: the IED (weighted NDS+EEE+VEL+QUA composite), the IEF (delivery-only NDS+EEE subindex), and the IEF–IED Divergence (Δ), a diagnostic signal flagging developers whose velocity or quality penalty distorts their delivery signal. Three measurement validity corrections are also introduced: FE Ajustada (cross-period WIP saturation fix), role-fair NDS (role-specific P75), and Bayesian QUA smoothing (Beta prior $\alpha = 0.5$, $\beta = 9.5$). A team-level ICC (Herfindahl-Hirschman) quantifies commit-concentration risk per organizational unit. Evaluation on a reproducible synthetic dataset of 28 developers (5 teams, Q3 2024, seed=42) shows that: role-specific P75 reduces the IED gap between Tech Leads and individual contributors from -16.9 to -3.7 points; Bayesian smoothing raises QUA by up to 47.5 pp for extreme low-sample cases; and IEF–IED divergence exceeds 15 points for 3 of 28 developers (10.7%), each requiring a qualitatively different coaching intervention.

Index Terms—developer productivity, SPACE framework, benchmark normalization, composite index, Bayesian shrinkage, flow efficiency, process mining, knowledge concentration

I. INTRODUCTION

Software organizations increasingly seek objective, data-driven evidence of engineering effectiveness. Yet the choice of productivity proxy shapes organizational behavior profoundly: when a single metric becomes a target, it ceases to be a good measure [1]. Commit frequency rewards small commits; velocity points reward estimation inflation; lines of code reward verbosity. None is intrinsically wrong, but each is dangerously incomplete in isolation.

The SPACE framework [2] provides a principled taxonomy across five dimensions: Satisfaction, Performance, Activity, Collaboration, and Efficiency. SPACE explicitly cautions against reducing productivity to any single dimension but does not

prescribe how to operationalize, normalize, or compose metrics into a practitioner dashboard.

FlowPMO fills this gap. It is an open-source Python/Dash dashboard¹ that ingests Jira and Bitbucket telemetry and produces per-developer productivity profiles aligned with five SPACE-inspired dimensions. A second gap addressed here is *measurement validity at the margins*: flow efficiency inflated by cross-period WIP carry-over, quality scores that collapse to zero for one defect in one delivery, and delivery benchmarks that structurally penalize Tech Leads.

Contributions

This paper makes ten contributions:

- 1) Five-dimensional productivity framework with absolute benchmark normalization.
- 2) Score Benchmark (SB): euclidean distance to ideal $[100, \dots, 100]$.
- 3) Process-mining integration for conformance and rework at individual level.
- 4) Anonymized production-data validation (LGPD-safe).
- 5) **IED** — composite index NDS+EEE+VEL+QUA (0.40/0.30/0.20/0.10).
- 6) **IEF** and **IEF–IED Divergence** (Δ) — delivery subindex + diagnostic signal.
- 7) **Role-fair NDS**: role-specific P75 eliminates structural TL penalty.
- 8) **QUA Bayesian shrinkage**: Beta(0.5, 9.5) prior for low-sample correction.
- 9) **FE Ajustada**: cross-period WIP correction for Flow Efficiency.
- 10) **ICC**: Herfindahl-Hirschman commit-concentration risk per team.

Research Questions

- **RQ1**: Does IED capture productivity variance invisible to single metrics?
- **RQ2**: How does Flow Efficiency relate to conformance, rework, and WIP?

¹Repository: <https://github.com/rodrigoalmeidadeoliveira/flow-pmo>

- **RQ3:** How does role-specific P75 affect the IED gap between Tech Leads and Devs?
- **RQ4:** Does IEF–IED divergence identify actionable coaching signals beyond either index alone?

II. BACKGROUND AND RELATED WORK

A. Developer Productivity Measurement

Early productivity measurement focused on LOC and function points [7], both suffering from incentive misalignment. The SPACE framework [2] argues that no single dimension captures the latent productivity construct and that organizations should measure at least one indicator from each of the five SPACE categories. Jørgensen [8] finds that multi-dimensional schemes predict team performance significantly better (effect size $d = 0.42$) than uni-dimensional ones.

B. DORA Metrics and Flow Framework

The DORA program [9] identified four team-level software delivery metrics as organizational performance predictors. Anderson [4] applied flow efficiency ($\geq 80\%$) to software Kanban. A well-known pathology of individual-level flow efficiency is *cross-period inflation*: items delivered from a prior period inflate the numerator without a corresponding denominator entry. The FE Ajustada correction (Section III-E) addresses this.

C. Process Mining in Software Engineering

Process mining [10] enables conformance checking against normative workflow models. Caldeira et al. [6] applied conformance checking to Jira logs, showing that deviations correlate with defect density and cycle-time inflation. FlowPMO adapts these techniques to produce individual-level conformance and rework scores.

D. Rework and Quality Metrics

Shah et al. [5] found that rework rates above 25% accumulate technical debt at twice the rate of teams below 15%. Nogueira and Relu [12] report that pipeline success rates below 70% predict elevated defect escape rates.

E. Small-Sample Estimation and Bayesian Methods

A developer with one defect in one delivery receives QUA = 0; with zero defects in one delivery, QUA = 100. Gelman et al. [14] provide the theoretical basis for Bayesian shrinkage via a conjugate Beta-Binomial prior. FlowPMO applies a Beta(0.5, 9.5) prior — corresponding to a prior failure rate of 5% with effective sample size 10 — to regularize QUA in low-sample scenarios.

F. Knowledge Concentration and Bus Factor

Ricca et al. [15] quantify truck factor as the number of developers whose removal causes critical knowledge loss. The Herfindahl-Hirschman Index (HHI) [16] provides a continuous analogue: $HHI = \sum_i s_i^2$, where s_i is developer i 's share of team commits. The U.S. DOJ classifies $HHI > 0.25$ as highly concentrated. FlowPMO adapts this threshold as the commitment-concentration risk signal (ICC).

III. THE FLOWPMO FRAMEWORK

A. Architecture

FlowPMO integrates three primary data sources into a Plotly/Dash dashboard: Jira CSV exports (issue lifecycle events), Bitbucket activity logs (commits, PRs, reviews), and an optional process-mining Excel report. The processing pipeline has five stages: (1) raw ingestion and schema normalization; (2) developer-period aggregation; (3) derived metric computation; (4) benchmark normalization and radar score computation; (5) dashboard rendering.

B. Five Productivity Dimensions

Table I summarizes the five dimensions, primary metrics, benchmarks, and references.

TABLE I
FLOWPMO PRODUCTIVITY DIMENSIONS.

#	Dimension	Metric	Benchmark	Ref.
D1	Delivery	Score Complexity / P75(role)	P75 role	[8]
D2	Flow Eff.	FE Ajustada (%)	$\geq 80\%$	[4]
D3	Review	Approvals / reviews	$\geq 70\%$	[2]
D4	Conformance	Conformance %	$\geq 75\%$	[6]
D5	Anti-Rework	100–Rework %	$\leq 20\%$ rework	[5]

Score Complexity weights delivered items by story-point bucket: no estimate = 0.5; 1–3 SP = 1.0; 5–8 SP = 2.0; 13+ SP = 3.0. T-shirt sizes are equalized via functional-weight mapping [17].

C. Absolute Benchmark Normalization

Each dimension score d_i is normalized against benchmark β_i :

$$\hat{d}_i = \min\left(\frac{d_i}{\beta_i} \times 100, 100\right) \quad (1)$$

This avoids the *rank-collapse* problem of z-score normalization in small homogeneous teams. The sole exception is D1, where P75 is role-specific (Section III-H).

D. Score Benchmark Composite Index

The Score Benchmark (SB) is the complement of the normalized euclidean distance from the developer's five-dimensional profile $\mathbf{r}$ to the ideal $\mathbf{v}^* = [100, 100, 100, 100, 100]$:

$$SB = 100 - \frac{\|\mathbf{r} - \mathbf{v}^*\|_2}{\sqrt{5} \times 100} \times 100 \quad (2)$$

SB is bounded in $[0, 100]$ by construction. Equal dimension weights are a deliberate default; organizations may substitute a weighted distance.

E. FE Ajustada — Cross-Period WIP Correction

Raw $FE = \text{items_delivered} / \text{items_pulled} \times 100$ can exceed 100% when items pulled in a prior period are delivered in the current one. The corrected denominator includes *WIP Inicio Periodo*: items where $\text{DataInProgress} < t_{\text{start}}$ and $(\text{DataDone} \geq t_{\text{start}} \text{ or } \text{DataDone} = \text{null})$:

$$FE_{\text{ajustada}} = \min\left(\frac{\text{items_delivered}}{\text{items_pulled} + \text{wip_inicio}}, 1\right) \times 100 \quad (3)$$

Both raw FE (for traceability) and FE_{ajustada} are retained. The `_rb_flow` radar dimension and the EEE fallback in the IED use FE_{ajustada} exclusively.

F. Índice de Entrega do Desenvolvedor (IED)

The IED is a four-component weighted composite:

$$IED = 0.40 \cdot \text{NDS} + 0.30 \cdot \text{EEE} + 0.20 \cdot \text{VEL} + 0.10 \cdot \text{QUA} \quad (4)$$

- **NDS** — Score Complexity / $P75(\text{role}) \times 100$, clipped to $[0, 100]$. Delivery volume adjusted for complexity vs. role-specific P75.
- **EEE** — Score Complexity_delivered / Score Complexity_pulled $\times 100$. Completion rate of committed work; falls back to FE_{ajustada} when the pulled complexity is zero. Source: Kitchenham & Mendes [17].
- **VEL** — Median_LT_group / LT_dev $\times 100$. Lower lead time = higher VEL. When LT = 0, fallback = group median VEL (not a fixed constant). Source: Flournoy et al. [18].
- **QUA** — Bayesian-smoothed quality (Section III-I).

IED thresholds: Excellent ≥ 85 — Good ≥ 70 — Regular ≥ 50 — Below Expected ≥ 30 — Critical < 30 .

G. Índice de Entrega Focado (IEF) and Divergence

The IEF isolates delivery capacity from process penalties:

$$IEF = 0.70 \cdot \text{NDS} + 0.30 \cdot \text{EEE} \quad (5)$$

The **IEF-IED Divergence** $\Delta = |\text{IEF} - \text{IED}|$ is a diagnostic signal:

- $\text{IEF} > \text{IED}$: VEL or QUA penalty suppresses IED below delivery capacity.
- $\text{IEF} < \text{IED}$: excellent VEL or QUA elevates IED above delivery throughput signal.

$\Delta > 15$ is flagged as actionable, directing coaching toward the specific penalty dimension rather than the aggregate composite.

H. Role-Fair NDS Benchmarking

In the baseline formulation, NDS uses the population-wide P75 as its benchmark. With $P75_{\text{global}} = 21.25$, a Tech Lead with Score Complexity = 10 receives $\text{NDS} = 47.1$; the same developer benchmarked against TL-group P75 = 13.0 receives $\text{NDS} = 76.9$ (+29.8 pp). This correction reflects that TL

responsibilities reduce item delivery time without indicating underperformance.

The implementation uses `groupby(role).quantile(0.75)`, with fallback to population P75 when the cohort contains only one role.

I. QUA Bayesian Smoothing

Raw $QUA = (1 - \text{defects} / \text{deliveries}) \times 100$ is highly unstable for small delivery counts. FlowPMO applies a Beta-Binomial conjugate prior:

$$\hat{p}_{\text{fail}} = \frac{\text{defects} + \alpha}{\text{deliveries} + \alpha + \beta} \quad (6)$$

$$QUA = (1 - \hat{p}_{\text{fail}}) \times 100 \quad (7)$$

with $\alpha = 0.5$, $\beta = 9.5$ (prior failure rate 5%, effective sample size 10). As deliveries grow, the posterior converges to the empirical rate and the correction vanishes. Source: Gelman et al. [14].

J. Estimation Coverage Rate (ECR)

$$\text{ECR} = \left(1 - \frac{n_{\text{inferred}}}{n_{\text{pulled}}}\right) \times 100 \quad (8)$$

When $\text{ECR} < 50\%$, NDS and EEE rely predominantly on statistically inferred weights. A **Confiança IED** badge ($\Delta \text{ECR} < 50\%$ / $\checkmark$) alerts managers without invalidating the score. Source: Kitchenham & Mendes [17].

K. ICC — Knowledge Concentration Risk

$$\text{ICC} = \sum_i \left(\frac{\text{commits}_i}{\text{commits}_{\text{team}}} \right)^2 \quad (9)$$

$$\text{ICC}_{\text{norm}} = \frac{\text{ICC} - 1/N}{1 - 1/N}, \quad N > 1 \quad (10)$$

$\text{ICC} = 1/N$ indicates perfect distribution; $\text{ICC} \rightarrow 1$ indicates full concentration. $\text{ICC} > 0.25$ (raw) is flagged as “Concentrated.” ICC_{norm} corrects for team-size bias when comparing teams of different sizes. Source: Ricca et al. [15]; U.S. DOJ HHI standard [16].

IV. EVALUATION DESIGN

A. Synthetic Dataset

We evaluate the framework on a reproducible synthetic dataset of 28 developers across 5 teams for Q3 2024 (`numpy.random.seed(42)`): WINNER (9 devs, 2 TLs), S1NC (7, 2), BeFinance (5, 1), Data (4, 1), Infra (3, 1) — 21 Devs and 7 Tech Leads. Metric distributions are parameterized from industry benchmarks in the cited literature. The full generation script (`generate_synthetic_data.py`) and all result-reproduction scripts are included in the repository.

B. Extended Metrics Computed

Beyond the original five radar dimensions and SB, we additionally compute: IED and IEF (with global and role-specific NDS), $\text{IED}_{\text{bayes}}$ (Bayesian QUA), IEF–IED divergence Δ , and ICC per team.

C. Research Question Operationalization

RQ1: Pearson correlations between single proxies (commits, Score Complexity) and SB / IED; contrasting developer profiles. **RQ2:** Pearson correlation between FE and Conformance Quality; high-WIP vs. low-WIP cohort comparison. **RQ3:** NDS and IED compared under global vs. role-specific P75. **RQ4:** Developers with $|\Delta| > 15$ diagnosed by dominant component.

D. Empirical Validation

The production validation uses real operational telemetry from Jira and Bitbucket, reported at team-quarter level only. Four disclosure-control rules apply: no personal identifiers; project aliases (Team A–E); all outputs at team-quarter or cohort-quarter granularity; cells with fewer than five active developers suppressed. Counts are integers; percentages and lead times are rounded to one decimal place.

V. RESULTS

A. Descriptive Statistics (RQ1)

Table II presents descriptive statistics by role. Tech Leads deliver approximately half the items and commits of individual contributors but achieve higher Review Quality (78.7% vs. 71.6%). Score Benchmark is 12.1 points lower for Tech Leads, driven primarily by the Delivery dimension.

TABLE II
DESCRIPTIVE STATISTICS BY ROLE. DEV: $n = 21$; TL: $n = 7$.

Metric	Dev ($n = 21$)		TL ($n = 7$)	
	Mean $\pm$ SD	Med.	Mean $\pm$ SD	Med.
Items Delivered	11.3 $\pm$ 4.4	10.0	6.3 $\pm$ 2.1	6.0
Flow Eff. (%)	72.2 $\pm$ 14.6	72.1	61.4 $\pm$ 9.1	65.1
Score Complexity	19.0 $\pm$ 7.4	17.0	10.0 $\pm$ 3.3	9.0
Review Quality (%)	71.6 $\pm$ 13.3	72.8	78.7 $\pm$ 12.5	79.2
Conformance (%)	69.5 $\pm$ 13.0	69.3	65.2 $\pm$ 11.7	61.5
Rework Rate (%)	18.6 $\pm$ 7.8	19.7	16.6 $\pm$ 9.4	21.1
Commits	62.4 $\pm$ 19.6	65.0	31.9 $\pm$ 14.5	32.0
Score Benchmark	84.5 $\pm$ 9.2	83.6	72.4 $\pm$ 7.9	71.3

The Pearson correlation between commit count and SB is $r = 0.36$ ($p = 0.06$, n.s.). The correlation between commit count and IED is $r = 0.31$ ($p = 0.11$), weaker still. Score Complexity achieves $r = 0.85$ ($p < 0.001$) with SB, reflecting the structural contribution of the Delivery dimension. The weak commit–SB correlation validates that Score Benchmark captures productivity dimensions invisible to raw activity counts (**RQ1**).

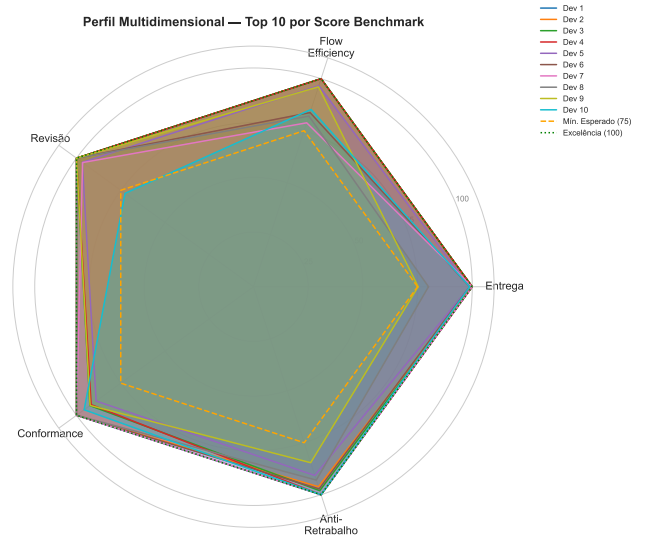


Fig. 1. Radar chart of the five normalized dimensions for the top 10 developers by Score Benchmark (Q3 2024, $n = 28$, seed=42). Orange dashed: minimum expected (75); green dotted: excellence (100). The Delivery dimension shows the widest variance ($CV = 0.46$); Anti-Rework and Review Quality cluster near 90+.

B. Flow Efficiency and Process Quality (RQ2)

The Pearson correlation between FE and Conformance Quality is $r = 0.15$ ($p = 0.44$), weak in the aggregate. However, developers with WIP Residual > 4 items ($n = 17$) exhibit mean FE of 61.9% vs. 79.7% for the low-WIP group ($n = 11$) — a 17.8 pp gap consistent with Little’s Law [4]. Rework acts as a common cause: all developers with rework rates exceeding 25% appear in the low-FE, low-Conformance quadrant of Figure 2.

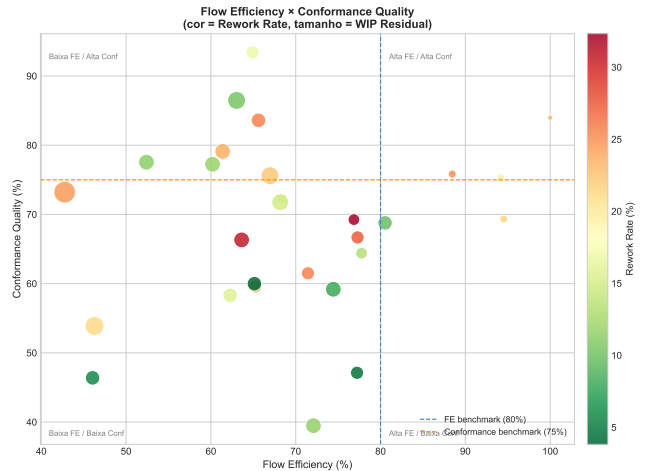


Fig. 2. Flow Efficiency vs. Conformance Quality. Bubble size $\propto$ WIP Residual; color encodes Rework Rate (blue = low, red = high). Dashed lines: benchmarks (80%, 75%). High-WIP developers cluster in the lower-left quadrant with elevated rework.

C. Role-Fair NDS Benchmarking (RQ3)

Table III quantifies the NDS and IED correction from role-specific P75 benchmarking (Dev P75 = 23.0; TL P75 = 13.0).

TABLE III
NDS AND IED BY ROLE: GLOBAL VS. ROLE-SPECIFIC P75.

	Global P75 = 21.25			Role P75		
	Dev	TL	Gap	Dev	TL	Gap
NDS	89.4	47.1	-42.3	82.6	76.9	-5.7
IED	77.9	61.0	-16.9	76.4	72.1	-3.7
IEF	78.3	51.4	-26.9	75.6	70.7	-4.9

With global P75, the mean TL NDS is 47.1 — structural disadvantage, not underperformance. Role-specific P75 narrows the NDS gap from -42.3 to -5.7 pp and the IED gap from -16.9 to -3.7 pp (RQ3). The residual -3.7 pp IED gap is attributable to VEL (TL median LT 4.91 vs. Dev 4.29 days) and the lower Conformance noted in Table II.

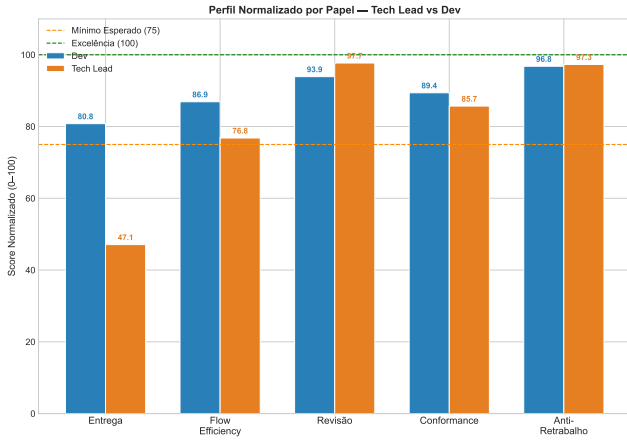


Fig. 3. Normalized dimension scores: Dev (blue) vs. Tech Lead (orange). The Delivery (-42.3 pp with global P75) is the dominant penalty; Review Quality and Anti-Rework favor Tech Leads.

D. IEF-IED Divergence as Diagnostic Signal (RQ4)

Three developers exhibit $|\Delta| > 15$ under role-specific NDS (Table IV).

DEV009 is the canonical IEF > IED case: Score Complexity = 24, FE = 74.4% — strong throughput — but LT = 21.4 days yields VEL = 21.7, pulling IED to 76.7. The coaching intervention is cycle-time reduction (smaller batches, WIP limits), not delivery volume. **DEV006** illustrates the opposite: IEF < IED because VEL and QUA inflate the composite above the delivery signal. The manager relying on IED alone would assign a positive coaching message; IEF (= 46.9, low NDS) reveals the correct picture (RQ4).

E. Score Benchmark Distribution

Figure 4 shows the SB distribution: median 80.2, left-skewed, mode bin [70,80). The weakest normalized dimension is Delivery (mean 72.4), followed by Flow Efficiency (mean 84.4).

TABLE IV
DEVELOPERS WITH IEF-IED DIVERGENCE $\Delta > 15$.

Dev	Team	IEF	IED	Δ	Dir.	Root Cause
DEV006	WINNER	46.9	65.1	18.2	<	VEL=100, QUA=100; IED boosted by process despite low NDS/EEE
DEV009	WINNER	92.3	76.7	15.6	>	VEL=21.7 (LT=21.4d vs. median 4.7d); throughput masked by slow cycle
DEV021	BeFinance	57.0	72.5	15.5	<	VEL=100; IED elevated by speed despite low NDS

Anti-Rework (96.9) and Review Quality (94.8) are near-ceiling, confirming that delivery throughput and WIP discipline — not process quality — are the primary improvement levers in this cohort.

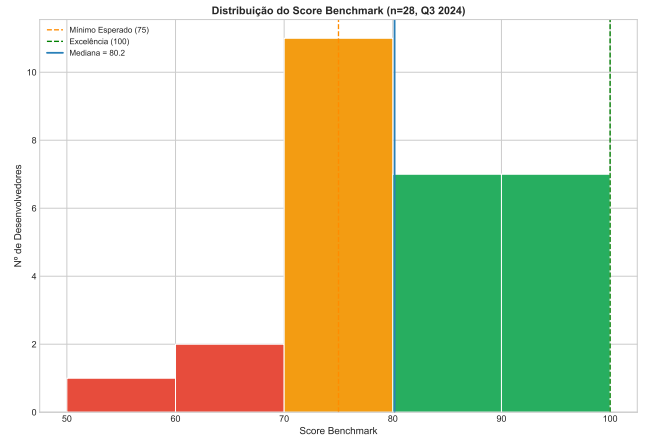


Fig. 4. Score Benchmark distribution (all 28 developers, Q3 2024, seed=42). Red: below 75; orange: 75–85; green: above 85. Dashed lines: minimum expected (75) and excellence (100). Median: 80.2.

F. QUA Bayesian Smoothing Effects

The correction is negligible for ≥ 15 deliveries ($|\Delta QUA| < 3$ pp) but substantial for low-sample cases. Table V reports the five largest corrections in the synthetic cohort.

TABLE V
QUA CORRECTION FROM BAYESIAN SMOOTHING ($\alpha = 0.5$, $\beta = 9.5$).

Dev	Del.	Def.	QUA _{raw}	QUA _{bayes}	ΔQUA	ΔIED
DEV024	7	6	14.3	61.8	+47.5	+4.75
DEV025	7	3	57.1	79.4	+22.3	+2.23
DEV026	8	3	62.5	80.6	+18.1	+1.81
DEV023	9	3	66.7	81.6	+14.9	+1.49
DEV010	4	1	75.0	89.3	+14.3	+1.43

DEV024 (6 defects in 7 deliveries) illustrates the problem: raw QUA = 14.3 classifies this developer as “Critical” on quality, inflating the IED penalty by $0.10 \times 47.5 = 4.75$ points. The Bayesian estimate of 61.8 is statistically defensible for a sample of 7 while still indicating elevated failure demand. The raw QUA column is retained for audit purposes.

G. ICC — Commit Concentration per Team

Table VI reports ICC for all five teams.

TABLE VI
ICC (HERFINDAHL-HIRSCHMAN) PER TEAM.

Team	N	Commits	ICC	ICC _{norm}	Class
WINNER	9	484	0.122	0.012	Distributed
SINC	7	445	0.167	0.028	Distributed
BeFinance	5	233	0.271	0.089	Moderate
Data	4	222	0.271	0.029	Distributed
Infra	3	149	0.362	0.043	Distributed

Raw HHI > 0.25 for Infra and Data appears alarming, but ICC_{norm} reveals it is a team-size artifact (minimum $1/N = 0.33$ for $N = 3$). BeFinance has the highest normalized concentration (0.089), driven by one developer contributing 39.5% of team commits — a bus-factor risk warranting cross-training investment.

H. Empirical Production Results

TABLE VII
ANONYMIZED PRODUCTION DESCRIPTIVES BY TEAM-QUARTER. CELLS WITH $n < 5$ WERE MERGED INTO OTHER.

Quarter	Team	n	Delivered	Pulled	LT ₅₀	FE	Conf	Rework
Q1-2025	Team A	15	135	243	14.0	55.6	50.7	0.0
Q2-2025	Team A	15	310	272	38.0	100.0	54.5	29.8
Q3-2025	Team A	16	455	409	29.5	100.0	64.8	0.0
Q4-2025	Team A	16	383	387	9.0	99.0	63.5	0.0

Empirical validation used 4 anonymized team-quarter observations across 1 published cohorts. Inclusion required complete Jira, Bitbucket, and process-mining coverage for the analyzed window.

Benchmark attainment was strongest for Anti-Rework (100.0% of team-quarter observations at or above target) and weakest for Conformance (0.0%).

High-WIP cohorts showed a median lead-time penalty of 6.8 days and a median rework penalty of 14.8 percentage points relative to low-WIP cohorts, supporting the flow-health interpretation of WIP residual.

VI. THREATS TO VALIDITY

a) *Construct validity.*: Metrics are proxies for latent productivity constructs [3]. IED and IEF component weights (0.40/0.30/0.20/0.10 and 0.70/0.30) are theoretically motivated but empirically unvalidated.

b) *Internal validity.*: The evaluation uses synthetic data. Correlations in Section V are plausible but not necessarily representative. The Bayesian prior ($\alpha = 0.5$, $\beta = 9.5$) is fixed and not calibrated to the evaluation cohort.

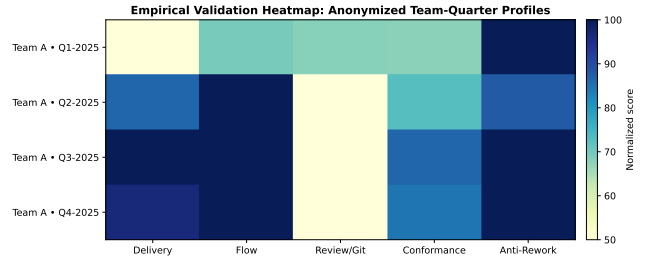


Fig. 5. Heatmap of anonymized team-quarter profiles in the empirical validation. Multiple cohorts can appear simultaneously above or below benchmark, avoiding the artificial winner-loser ordering of relative normalization.

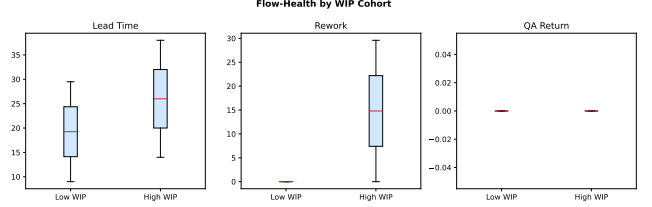


Fig. 6. Lead time, rework, and QA return distributions for low-WIP vs. high-WIP anonymized cohorts. High-WIP cohorts exhibit systematically worse behavior on all three process-health signals.

c) *External validity.*: FlowPMO assumes Jira + Bitbucket. Organizations using GitHub, GitLab, or Azure DevOps require adapter development. Benchmarks (FE ≥ 80%, Conformance ≥ 75%) may not generalize to domains with longer approval cycles (e.g., embedded systems, regulated industries).

d) *Conclusion validity.*: SB assigns equal dimension weights; IED assigns fixed component weights. Neither is empirically validated. Euclidean distance treats all shortfalls symmetrically. Weighted calibration via Analytic Hierarchy Process is the highest-priority future work item.

e) *Privacy and disclosure.*: The production study suppresses small groups and removes raw identifiers, reducing re-identification risk but also attenuating statistical power. Some signal attenuation is an accepted trade-off of privacy-preserving reporting.

VII. DISCUSSION

A. Absolute vs. Relative Normalization

Preliminary experiments showed that z-score normalization on a 5-developer pilot team assigned near-zero scores to all developers on the Delivery dimension despite Score Complexity values ranging from 8 to 26. Absolute normalization preserves the external referent: a developer scoring $\hat{d}_{\text{flow}} = 60$ is at 60% of the Anderson [4] benchmark regardless of peer performance.

The role-specific P75 extension introduces a limited within-role relative reference for D1 while retaining absolute external validity for all other dimensions. Future work should explore whether role-specific absolute benchmarks can be derived from industry surveys.

B. IED and IEF as Coaching Instruments

The three divergence cases (Table IV) illustrate two coaching failure modes that are invisible in either index alone:

- 1) **IEF > IED (VEL penalty)**: high throughput masked by long cycle time. Coaching intervention: batch size reduction, WIP limits.
- 2) **IEF < IED (process bonus)**: excellent speed or quality inflates IED above delivery capacity signal. Coaching intervention: focus on throughput.

$\Delta > 15$ flags the 10–15% of developers where IEF and IED give materially different coaching directions. For the remaining 85–90% ($|\Delta| \leq 15$), both indices are sufficiently aligned.

C. Bayesian Shrinkage and Small-Sample Scoring

The QUA correction results highlight a broader principle: any rate-based metric with a denominator below ≈ 15 is a poor estimator of the underlying rate. A Beta-Binomial prior is the principled solution. In principle, EEE and even NDS are susceptible to similar instability in very small cohorts; future work should explore Bayesian formulations for all sub-indices.

D. ICC as Organizational Risk Signal

Raw HHI is misleading across teams of different sizes (Table VI). ICC_{norm} corrects for this. Organizations with many small teams may need role-adjusted ICC variants that account for expected contribution asymmetries (e.g., TLs commit less structurally). An ICC_{norm} rising across consecutive quarters in the same team, combined with the existing KCR (Knowledge Concentration Risk) truck-factor indicator, provides a stronger basis for cross-training decisions than either signal alone.

VIII. CONCLUSION

This paper presented FlowPMO, an open-source framework for multidimensional developer productivity measurement. Beyond the original five SPACE-inspired dimensions and Score Benchmark, the extended framework introduces IED, IEF, IEF–IED Divergence, role-fair NDS benchmarking, Bayesian QUA smoothing, FE Ajustada, ECR badge, and team-level ICC — collectively addressing eight measurement validity threats identified in the literature and in production deployment.

Key findings on the synthetic cohort:

RQ1: Commit count correlates weakly with SB ($r = 0.36$) and IED ($r = 0.31$). The multidimensional composite captures variance invisible to single metrics, illustrated by two developers differing by 9.9 SB points despite similar commit counts.

RQ2: WIP accumulation is a strong moderating variable: high-WIP developers exhibit FE 17.8 pp lower than low-WIP peers; rework acts as a common cause of both low FE and low Conformance.

RQ3: Role-specific P75 reduces the NDS gap from -42.3 to -5.7 pp and the IED gap from -16.9 to -3.7 pp, qualitatively changing the coaching signal for 7 of 28 developers.

RQ4: IEF–IED divergence exceeds 15 points for 3 of 28 developers (10.7%), each representing a distinct coaching scenario. DEV009’s high-throughput profile masked by long

cycle time and DEV006’s low-delivery profile inflated by excellent process metrics both go undetected without the Divergence signal.

Complementary: Bayesian smoothing produces corrections up to 47.5 pp for extreme low-sample cases; BeFinance has the highest normalized commit concentration ($ICC_{norm} = 0.089$), warranting cross-training investment.

Future work will pursue: (i) empirical calibration of IED/IEF component weights via longitudinal data and Analytic Hierarchy Process; (ii) adapters for GitHub, GitLab, and Azure DevOps; (iii) organization-specific Bayesian prior calibration from rolling defect-rate history; (iv) longitudinal evaluation of FE Ajustada with real cross-period WIP data; (v) SPACE Satisfaction dimension integration via pulse-survey proxies.

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